

TerraFly: Depth Wizard

Single-View Metric Height Estimation & Cinematic 3D Flythrough Synthesis

PROBLEM STATEMENT

Single View Height Estimation & 3D Flythrough

CORE TECHNOLOGY

Depth ViTs, 3D Ray Unprojection, Trajectory Synthesis

TARGET SECTORS

Disaster Response, Defense Recon, Urban Planning

1 Introduction: Problem & Solution

PRESENTER SPOKEN SCRIPT (30 SECONDS)

"Respected Judges, in critical missions like disaster relief, military reconnaissance, or urgent infrastructure audits, responders often only possess a single 2D aerial snapshot or satellite photo. Today, estimating height or understanding the 3D terrain requires expensive multi-angle drone flights or LiDAR hardware taking hours to compute. We present **TerraFly** — an AI system that takes just **one single 2D image** and instantly reconstructs its 3D depth, estimates building and terrain heights in meters, and creates interactive cinematic flythrough videos in under 5 seconds."

The Critical Bottleneck

- **LiDAR Scanners:** Prohibitively expensive (\$10k-\$100k) & heavy.
- **Photogrammetry (SfM):** Demands 50+ overlapping images & high compute.
- **Zero 3D from Single Photos:** Conventional GIS cannot extract 3D elevations from single legacy or reconnaissance photos.

The TerraFly Breakthrough

- **Zero-Shot Monocular 3D:** Generates dense metric 3D point clouds from 1 photo.
- **Metric Height Measurement:** Live calibrated caliper measuring any building/cliff.
- **Cinematic Flythroughs:** Generates 360° orbit, zoom, and terrain flyby videos.

2 Technical Approach & Architecture

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*"Our architecture combines state-of-the-art Vision Transformers with geometric ray casting. First, our **Monocular Depth Engine** extracts high-frequency structural depth boundaries. Second, our **Metric Height Calibration module** fits a ground plane equation to calculate precise relative heights in meters. Third, we unproject the 2D pixels into a 3D coordinate space and compute smooth parametric Bézier camera trajectories to render cinematic flythrough videos with occlusion handling — all exposed through an interactive real-time Three.js interface."*

1. Monocular Depth ViT Backbone

Utilizes Vision Transformer backbones (Depth-Anything / MiDaS) with multi-scale feature aggregation to predict dense relative depth maps with sharp object boundaries.

2. Metric Calibration & Ground Plane

RANSAC-based ground plane detection segments terrestrial base levels from vertical structures, converting relative disparity into real-world metric elevation (meters).

3. 3D Camera Trajectory Engine

Generates smooth 6-DOF orbital, spiral, and forward flythrough paths with depth-guided mesh inpainting to eliminate visual occlusion artifacts.

4. Real-Time Web Stack

High-performance FastAPI backend with PyTorch acceleration paired with a responsive Three.js WebGL viewport for live point-cloud manipulation and export.

3 Feasibility, Challenges & Risk Mitigation

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"The biggest challenge in single-view 3D is scale ambiguity and hidden occlusions when moving the camera. We overcame this by implementing ground-plane anchoring for metric scaling, and depth-aware mesh blending for smooth trajectory video rendering. TerraFly is engineered to run seamlessly on standard consumer hardware and edge devices without needing internet connectivity or cloud GPUs."

Technical Challenge	Risk Factor	TerraFly Mitigation Strategy
Scale Ambiguity	Monocular images lack absolute physical distance markers.	Reference height calibration & ground-plane regression algorithm.
Occlusion Holes	Novel view flythroughs reveal unseen surfaces behind buildings.	Depth-aware edge clamping & adaptive mesh point interpolation.
Hardware Constraints	Heavy deep learning models cannot run in tactical field setups.	FP16 half-precision, model quantization, and CPU-fallback pipeline.

4 Impacts & Benefits

Defense & Tactical Operations

Enables instant 3D terrain elevation and building reconnaissance from a single surveillance UAV image without alerting adversaries or flying repeated patterns.

Disaster Relief & Flood Management

Rapidly calculates floodwater heights, bridge clearance levels, and landslide volume from standard rescue drone photographs.

Smart City & Infrastructure Audits

Automates urban height compliance and telecommunication tower line-of-sight analysis at a fraction of traditional survey costs.

95% Cost & Time Reduction

Reduces processing turnaround from 4 hours (photogrammetry) or days (LiDAR deployment) down to **under 5 seconds**.

5 Existing Approaches vs. What Makes Us Different

Capability / Metric	LiDAR Surveys	Photogrammetry (SfM)	NeRF / Gaussian Splat	TerraFly (Our Solution)
Input Required	Physical Laser Sensor	50 - 200 Images	40 - 100 Calibrated Photos	1 Single Photo (Any Camera)
Processing Time	Days (Post-processing)	30 mins - 4 hours	15 - 45 mins training	< 5 Seconds (Instant)
Hardware Cost	\$15,000 - \$100,000	High-End Workstations	High-VRAM GPU Required	\$0 (Consumer Laptop / Edge)
3D Flythrough Video	✗ Complex CAD setup	✗ Needs manual rigging	✓ High compute cost	✓ One-Click Auto Synthesis
Metric Height Tool	✓ Native point cloud	✓ Scale bar needed	✗ Relative radiance only	✓ Live Calibrated Caliper

6 Vision & Closing Pitch

Future Roadmap

Phase 1 (Current): Single-view metric depth, 3D point cloud generation, and trajectory flythrough video rendering.

Phase 2 (Next Quarter): Native integration with ISRO Cartosat / Bhuvan GIS satellite APIs for direct geographical coordinate matching.

Phase 3: Ultra-lightweight edge firmware for direct deployment on tactical defense and SAR micro-drones.

CLOSING STATEMENT FOR JUDGES (20 SECONDS)

*"To conclude: **TerraFly** bridges the gap between static 2D imagery and dynamic 3D spatial intelligence. We turn any single photograph into actionable metric heights and cinematic aerial flythroughs in seconds — empowering defense forces, disaster teams, and urban planners with real-time 3D vision at zero additional hardware cost. Thank you, and we are now ready for live demonstration and questions!"*